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 Studierichting: Informatica
 Oefeningenreeks 7B nr. 3.1

Kettingregel:

$$\left(\frac{\partial r}{\partial x} \frac{\partial r}{\partial y} \frac{\partial r}{\partial z} \right) \left(\frac{\partial r}{\partial u} \frac{\partial r}{\partial v} \frac{\partial r}{\partial w} \right) \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix}$$

u=yz:

$$\frac{\partial r}{\partial u} = e^{u+v+w}$$

$$\frac{\partial u}{\partial x} = 0 \text{ en } \frac{\partial u}{\partial y} = z \text{ en } \frac{\partial u}{\partial z} = y$$

v=xz:

$$\frac{\partial r}{\partial v} = e^{u+v+w}$$

$$\frac{\partial v}{\partial x} = z \text{ en } \frac{\partial v}{\partial y} = 0 \text{ en } \frac{\partial v}{\partial z} = x$$

w=xy:

$$\frac{\partial r}{\partial w} = e^{u+v+w}$$

$$\frac{\partial w}{\partial x} = y \text{ en } \frac{\partial w}{\partial y} = x \text{ en } \frac{\partial w}{\partial z} = 0$$

$$\Rightarrow \left(\frac{\partial r}{\partial x} \frac{\partial r}{\partial y} \frac{\partial r}{\partial z} \right) = (e^{xy+xz+yz} \ e^{xy+xz+yz} \ e^{xy+xz+yz}) \begin{pmatrix} 0 & z & y \\ z & 0 & x \\ y & x & 0 \end{pmatrix}$$

$$\frac{\partial r}{\partial x} = (y+z)(e^{xy+xz+yz})$$

$$\frac{\partial r}{\partial y} = (x+z)(e^{xy+xz+yz})$$

$$\frac{\partial r}{\partial z} = (x+y)(e^{xy+xz+yz})$$

References